

Soorej Santhoshkumar Nair

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EDUCATION

University of Colorado, Boulder

Master of Science in Data Science — GPA: 4.00

Aug 2025 – May 2027

Boulder, CO

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science

Aug 2017 – May 2021

Pilani, India

TECHNICAL SKILLS

Languages & Tools: Python, C++, C, Rust, Java, SQL, MATLAB, PowerBI, Tableau, Numpy, Pandas, Matplotlib

Frameworks: TensorFlow, Keras, Hugging Face, OpenCV, Scikit-learn, Linux, Databricks MLflow, Statistics

ML/AI: Diffusion Models, Vision-Language Models (CLIP, BLIP, Stable Diffusion), GANs, Autoencoders, LangChain, Agentic AI, RAGs, Image Processing, Map-reduce, Reinforcement learning, Machine Learning, Regression and Classification

Cloud & Tools: AWS, GCP, Docker, Kubernetes, Git, MLflow, Airflow, Google Cloud Platform, Hadoop, Spark

EXPERIENCE

Senior Data Scientist

Sep 2024 – Jul 2025

XPO Logistics

Pune, India

- Led data-driven product analytics and developed a Random Forest churn prediction model that achieved 90% precision in the top decile, preventing over \$240M in potential revenue loss. Translated quantitative insights into business recommendations that increased customer retention by 6%.
- Introduced an OCR-based computer vision system processing 60K+ bills of lading daily. Extracted structured data into BigQuery and SQL-based pipelines, supporting large-scale data measurement, metrics tracking, and evaluation frameworks.
- Designed an end-to-end LLM-based pipeline using Google Gemini, GCP, and Vertex AI integrating CV, NLP, and OCR workflows for automated invoice extraction and classification, enabling scalable, production-ready data science applications

Data Scientist

Jul 2022 – Aug 2024

XPO Logistics

Pune, India

- Partnered with McKinsey to develop ML-based logistics simulation model (Cerebro) implemented on 300+ centers, applying program evaluation, statistical experimentation, and quantitative analysis to improve freight handling efficiency by 54%.
- Developed a computer vision audit platform using GCP, Pub/Sub, SQL, and Vertex AI Pipelines, processing 80K+ documents daily, enhancing data quality metrics and saving 10K+ manual hours annually.
- Created CNN-based models for accessorial optimization and trailer image analysis. Analyzed statistical trends and performance metrics to improve product reliability and generate \$1.3M in incremental revenue.

Associate Data Scientist

Jul 2021 – Jun 2022

XPO Logistics

Hyderabad, India

- Built decision tree models to detect fraudulent claims using Vertex AI and SQL queries on structured datasets, generating \$15K monthly. Enhanced scalability, reproducibility, and statistical model evaluation.
- Developed ARIMA-based time series forecasting models in R, Python, and BQML to predict employee attrition. Communicated insights through data visualization, storytelling, and stakeholder reports for data-driven workforce planning.

PROJECTS

Multimodal Vision-Language Model (VLM) from Scratch

- Implemented a custom Vision-Language Model in PyTorch, building the core Transformer architecture (Multi-Head Attention, RoPE, KV-caching) and integrating a SigLIP-based vision encoder with a language model via a custom MLP projector for cross-modal alignment.
- Optimized training and inference by implementing causal masking and RMSNorm, improving stability and enabling efficient token generation for 7-DoF robotic action sequences.

Transformer based Hindi poetry generator

- Built a custom decoder only Transformer-based generative model leveraging natural language processing (NLP) and statistical text analysis on data fetched from web scraping.
- Applied data wrangling and model evaluation metrics to assess creativity and linguistic coherence.

Implementation of Sparsity Invariant CNNs

- Implemented sparse convolutional neural networks (SCNNs) for LiDAR data for vehicle driving LiDAR data, applying statistical validation and quantitative performance measurement to compare against traditional CNNs.